

# NTRO Bitcoin AML Surveillance

## Prototype Demo Guide

A plain walkthrough of the six parts of the prototype. Each part lists what it is and what every feature does — written to be read once, then used as a rehearsal reference.

RUNS AT  
**localhost:3000**  
6 PARTS

**What this system does.** It watches Bitcoin transaction traffic for NTRO and flags money-laundering activity — linking the network side (IP, ASN, geolocation) to the blockchain side (wallets, transactions), then using ML to group related wallets, spot unusual behaviour, catch laundering patterns and score risk. The whole platform runs **offline and air-gapped**.

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### Dashboard

The main screen and the starting point — system status, filters, and the live list of every alert.

#### Top Status Bar

AIR-GAPPED OPERATIONAL15,873 WALLETS INDEXED

The black bar across the very top. It carries the NTRO identity, the green Air-Gapped Operational status showing the system runs with zero internet, and a live count of how many wallets have been indexed. This is the first thing to point at when the demo opens.

#### Dual Transformer Chip

4.8ms CPU

A status pill for the AI engine showing combined inference latency on CPU. Clicking it opens the full benchmark and provenance audit — that is **Part 6**, and it is the strongest single credibility moment in the build.

#### Ingest Batch Button

The button in the top-right corner for uploading new transaction data. Clicking it opens the Ingest flow, which is **Part 5**. Worth mentioning early, so the audience knows the system is not frozen on a fixed demo dataset.

#### Stream Telemetry

The left-hand panel showing how much of the indexed data is currently in view. The progress bar shrinks as filters are applied and the percentage updates live, so the act of narrowing down is visible rather than hidden.

#### Risk Verdict Filters

CRITICAL

HIGH

MEDIUM

LOW

Four risk levels, each showing a live count of how many wallets sit in it. Clicking CRITICAL instantly cuts the table from roughly fifteen thousand rows down to the 103 most dangerous — the fastest way to show that triage actually works.

#### Min Anomaly Score

A slider for hiding anything scoring below a chosen threshold, so you keep only the most behaviourally unusual wallets. Four preset buttons let you jump straight to a useful range on stage without fiddling with the slider.

#### Laundering Heuristics

Peeling Chains

CoinJoin Mixing

Two switches for the deterministic laundering checks: Peeling Chains (linear multi-hop peel transfers) and CoinJoin Mixing (equal-output mixing rounds). Enabling both narrows the alert set down to wallets matching those two specific crime patterns.

#### Cluster Partition

A field for jumping directly to a numbered wallet group. Type an ID and the topology view loads that cluster — a fast way to move between a table finding and the graph that explains it.

#### View Tabs

Alerts Stream

Cluster Topology

The switch between the Alerts Stream list and the Cluster Topology graph. This is the main narrative spine of the demo: find a suspicious wallet in the list, then open that same entity in the graph to see who it connects to.

#### Alerts Table

One row per suspicious wallet, across seven columns: risk verdict with composite score, truncated address with one-click copy, anomaly score with percentile, cluster partition, the laundering pattern that fired, a known-seed flag, and the telemetry timestamp. Click any row to open its dossier.

#### Search & Live Status

Search wallets by address prefix — press / or Ctrl+K to jump to the field without touching the mouse. Rows load in batches as you scroll, and the footer carries a live entity counter and a ticking SYS UTC clock.

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# Drilling In — Dossier & Graph

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## NTRO Forensic Dossier

Opens when you click a wallet. The full evidence file for that one wallet — the core of the explainability story.

### Risk Gauge & Verdict

The headline number for this wallet: composite risk out of 1.0 and the band it falls into (CRITICAL, HIGH, MEDIUM, LOW). A red seed-family badge appears underneath when the wallet matches known threat intelligence.

### Automated Forensic Summary

A written paragraph in plain English explaining what pattern was found and why it matters. It is generated deterministically by the system, **not by an LLM**, so it can never invent or embellish a finding the way a chatbot could.

### Score Breakdown

The final score split into its four weighted parts — anomaly, risk, rule bonus and mixing, each shown to five decimal places. This is the direct answer to any "your score is a black box" challenge.

### Cluster Co-Spending Analysis

Which group the wallet belongs to, how large that group is, how close it sits to known bad wallets, and where it ranks on unusualness as a percentile. Concrete, quotable numbers rather than adjectives.

### Tabular Anomaly Check

How unusual this wallet's behaviour is versus normal traffic, shown as a reconstruction error, plus its percentile rank and whether it crossed the calibrated **0.0346** detection threshold.

### Mixing & Laundering Heuristics

The output of the deterministic laundering checks: mixing detected or not, chain length in hops, and where relevant a pass-through ratio showing how much value moved without changing hands.

### Triggered Rules Audit

Tags naming every individual rule that fired for this wallet, for example COINJOIN\_PATTERN\_DETECTED or LARGE\_CLUSTER. It lets you trace a verdict straight back to the exact rule that produced it.

### Explainability — SHAP Waterfall & Attention Matrix

Two views of the same evidence. The waterfall charts all **18 features** the model checked and which pushed this wallet towards being flagged; the heatmap shows how the model relates those features to each other, and **hovering a cell explains it in plain English**.

### Topological Subgraph & Export

The neighbouring wallets the model actually weighed, each with its own risk score, followed by the two export actions — **Download JSON** for the raw evidence, or **Export Legal Dossier** for the certified version covered in Part 4.

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## Cluster Topology

The network graph — how wallets, transactions and IP addresses connect. The visual payoff of the whole system.

### Force Graph Canvas

The main graph view. Wallets, transactions and IP hosts appear as nodes and their relationships as lines, with node colour driven by composite risk — **red is the most dangerous**. The stat bar above shows cluster ID, node count and edge count.

### Entity Filter Chips

**Seeds** **Wallets** **Txs** **IPs**

Toggle which node types are visible, each with a live count. Selecting **IPs** is a strong reveal because it exposes the shared infrastructure behind the wallets, not just the money trail.

### Search & Click-to-Inspect

Search an address and press Enter to fly the camera to that node. Clicking any node pins an inspector card showing its type, country, severity, risk score and degree, with a shortcut straight into the full dossier.

### Layout Modes

**Force** **Concentric** **Directed Flow**

Three arrangements of the same graph: Force Physics (default), Concentric Radar (nodes ringed by risk) and Directed Flow DAG (columns showing value flowing in and out). Switching between them live is one of the most impressive visual moments in the demo.

### Saliency Toggle & View Controls

Saliency highlights only the connections the transformer actually weighted as important, dimming the rest — a direct visual that the graph is driven by the model, not decoration. Alongside it: zoom, fit to screen, and pause physics to stop nodes drifting mid-sentence.

# The Closers — Legal, Ingest, Models

Legal · Data · Model

## 4 Section 65B & BSA 2023 Forensic Court Dossier

Turns a finding into a document a court will accept — the answer to "could you actually use this as evidence?"

### Official Layout & Statutory Basis

Rendered as a formal government certificate: NTRO letterhead, a classification marking, and the legal basis under both **Section 65B of the Indian Evidence Act 1872** and **Section 63 of the Bharatiya Sakshya Adhiniyam 2023**.

### Certificate ID & Statutory Declaration

A unique certificate reference, the jurisdiction, and the air-gapped device node the document was produced on — followed by the sworn declaration that the records came from this system, unaltered.

### Findings & Model Provenance

The target wallet's verdict, risk score, anomaly error, percentile rank and cluster in a formal findings table, plus SHA-256 checksums of the exact model weight files used to produce them.

### SHA-256 Seal

A tamper-evident fingerprint over the entire certificate. Alter a single character and the seal no longer validates — this is the strongest answer to "can this really survive as evidence?"

### Print / Save PDF

Exports a clean, self-contained printable certificate with the interface chrome stripped out. Ending the demo on a produced document is a strong closing beat for the judges.

## 5 Ingest

Uploading new transaction data — the answer to "what happens when fresh data arrives?"

### Drag & Drop Upload .CSV .JSON .XML

Accepts CSV, JSON and XML up to 500 MB, by drop or by browsing. The file is fingerprinted with a **SHA-256 hash** as it lands, before any parsing or processing begins.

### Sample Dataset Card

A pre-loaded 2,000-transaction stress-test file bundled with the app, which you can drag straight into the dropzone. It keeps the upload demo fully offline and removes any risk of hunting for a file on stage.

### Background Processing

A live progress bar while a Celery worker parses, validates, geo-enriches and bulk-loads the data. The upload is queued and processed asynchronously rather than blocking the browser.

### Duplicate Detection & Immediate Scoring

Re-uploading the same file is rejected as a duplicate, which is worth demonstrating on purpose to show the system is idempotent. Newly ingested wallets are also scored inline straight away, so it works on brand-new data without a full retrain.

## 6 Dual Transformer Model Provenance & Benchmark Audit

Opens from the blue chip in the top bar — the answer to "how do you know the AI is any good?"

### Four Headline Metrics

The four numbers that answer "is the model any good": graph F1 of **0.921** against a 0.880 promotion gate, **94.8%** peeling recall, **4.8 ms** combined inference latency on CPU, and **231 KB** of total model weights. Read all four aloud.

### Head-to-Head Comparison 5 dimensions

Our architecture against the standard baseline across five dimensions: the tabular anomaly engine, the graph proximity engine, edge relations modelled, explainability, and deployment footprint. It shows the improvement is measured rather than claimed.

### Model Weight Checksums

SHA-256 hashes of the exact weight files, with a copy button on each. These tie the numbers on screen to specific files, which is what reproducibility actually means in practice.

### Runs on a Basic Laptop

4.8 ms on CPU with 231 KB of weights and no GPU at all, cold booting in under 120 ms — that lightweight footprint is what makes the air-gapped field deployment claim credible.